CP003651.1:950300-963740
Thermococcus cleftensis

ATPase_2 HTH_5 DUF234 ZT_dimer Cation_efflux ? PUA_2 ATP-sulfurylase AF2212-like DHH DHHA1 Phosphodiest DUF4258 DUF2283 APS_kinase TauE Polysacc_synt MatE Sulfatase Sulfotransfer_3 Glycos_transf_2

AP018314.1:5089343-5105303
Sphaerospermopsis kisseleviana NIES-73

HTH_OrfB_IS605 STY-kinase DUF2267 ATP_transf Ap4A_phos_N Aminotran_5 Aminotran_5 SMC_N DUF2283 DUF4258 Phosphodiest Rubrerythrin HNH ? ? Heme_oxygenase HemN_C Radical_SAM

CCEJ010000011.1:95493-101217
Criblamydia sequanensis CRIB-18

Cation_efflux ZT_dimer ArsC PDZ_4 Peptidase_S41 ? ? ? DUF4853 DUF4258

LN885086.1:2777058-2790480
Candidatus Nitrospira inopinata

? ? DUF6754 ? MutL ? HsdM_N N6_Mtase DUF4258 MqsA_antitoxin ? Methylase_S Methylase_S Y1_Tnp AAA_14 DUF4143 ? Virulence_RhuM Fic

BK021730.1:8989-18364
Caudoviricetes sp.

Phage_terminase Terminase_2 Terminase_3 Terminase_3C SPP1_portal Phage_Mu_F DUF4258 Phage_scaffold ? ? ? ? ? ? ? ? ? HK97-gp10_like

BK026643.1:22076-30640
Caudoviricetes sp.

Terminase_2 Terminase_3 Terminase_3C SPP1_portal Phage_Mu_F DUF4258 Phage_scaffold ? ? ChrB_N ? ? BSP ? ? ? ?

BNWV01000003.1:35308-44645
Bacteroidia bacterium

HlyD_D23 ACR_tran Lip_A_acyltrans RteC PHM7_cyt ? ? DUF4258

DATAIZ010000220.1:12-1902
Geobacteraceae bacterium

Radical_SAM DUF2283 DUF4258 ?

DUMS01000037.1:57373-63225
Tissierellia bacterium

ABC_trans_CmpB Cu_amine_oxidN1 ? Terminase_2 ? Phage_min_cap2 NUMOD4 HNH_3 NUMOD1 Phage_min_cap2 DUF4258 ? Phage_gp49_66 ?

DUGS01000109.1:16436-36063
Methanosarcina sp.

NUDIX DUF3303 Acetylase LPG_synthase_C ThiC_Rad_SAM Toprim Topoisom_bac Zn_ribbon_Top1 Zn_ribbon_Top1 Ribosomal_S3Ae DUF4258 MqsA_antitoxin TPRs TPRs Hexapep RUBY_RBDX FeThRed_B Glutaredoxin MFS_1 MFS_1 FUR

DWYT01000052.1:24-12109
Candidatus Mediterraneibacter merdavium

AAA_22 Sirtuin DUF4258 ? Hydrolase E1-E2_ATPase HMA HTH_5 HTH_18 Peptidase_S11 DUF2871 Glyco_hydro_32C Glyco_hydro_32N

DATFNV010000188.1:266-11316
Gemmataceae bacterium

? DUF4258 DUF2283 HipA_2 DUF3037 Lactonase Phosphoesterase SOBP PFK FAD_binding_4 FAD-oxidase_C Fer4_8 CCG

DATFJL010000212.1:2933-9564
Balneolaceae bacterium

HNH_4 RraB Acyl-CoA_dh_2 23S_rRNA_IVP DUF6933 DUF6933 DUF2283 DUF4258 ACP_syn_III_C ACP_syn_III

DATDPE010000269.1:14906-32751
Anaerolineales bacterium

Sortase Sortase MFS_2 Guanylate_cyc AP-ATPase TPRs TPRs Guanylate_cyc AAA_22 TPR_MalT ? Transpeptidase Transgly DUF4258 DUF2283 Zn_ribbon_NUD Pterin_bind SpoIIAA-like ? RDD SpoIIM

DATGTL010000307.1:1-3894
Balneolaceae bacterium

? ? rve_3 rve HTH_Tnp_1 ? DUF4258

DAOOHF010000015.1:1-13603
Flavipsychrobacter sp.

PKD_4 PKD_4 CHU_C SMC_ScpB PA Peptidase_M28 PDZ_2 Phage_holin_4_2 ETRAMP DUF4258 FAD_binding_3 DHH DHHA1 RecJ_OB Peptidase_S66 Peptidase_S66C ? HlyD_D23 ACR_tran

DAORII010000070.1:5092-12842
Bacteroidia bacterium

PIN PhoH DUF4258 AlaDh_PNT_N AlaDh_PNT_C TsaE Receiver PglZ HD_assoc_2 HD Hexapep Hexapep Hexapep Hexapep LpxD Por_Secre_tail Por_Secre_tail Pur_ac_phosph_N

DAORHY010000003.1:6106-25386
Bacteroidia bacterium

DUF5777 DUF1223 Amidohydro_2 TPRs TPRs TMTC_DUF1736 PMT_2 3-HAO Apolipoprotein adh_short_C2 DUF4258 Aldedh DUF4412 TFIIE-A_C DUF885 AAA_33 TonB_dep_Rec_b-barrel Plug CarbopepD_reg_2 Por_Secre_tail PCMD SET

DATIHO010000088.1:7805-7628
Chlamydiales bacterium

DUF6088 AbiEii DLH ? TPRs DUF4258 DUF1016_N ? ? ? ?

DATLMY010000391.1:10-6139
Chitinophagaceae bacterium

Acyltransferase DUF4258 GMC_oxred_C GMC_oxred_N FAD_binding_2 Gluconate_2-dh3 ?

DASZWJ010000040.1:30135-39485
Gammaproteobacteria bacterium

Pox_A9 PriA_C Helicase_C Zn_ribbon_PriA DEAD PriA_3primeBD Rhodanese Glutaredoxin SecB NAD_Gly3P_dh_C NAD_Gly3P_dh_N ? PAAR_motif DUF4258 Phage_base_V Gp5_trimer_C Gp5_trimer_C

DATNOQ010001328.1:432-1689
Chitinophagaceae bacterium

Receiver DUF2283 DUF4258

WPAE01000015.1:22914-38406

Ignavibacteria bacterium

Trehalose_PPase Glyco_transf_20 pEK499_p136 MlaD ATG17_like ABC_tran MlaE MqsA_antitoxin DUF4258 UPF0102 RNase_HII DUF480 peroxidase peroxidase FUR BD-FAE Caps_assemb_Wzi

AKBK01000051.1:223-903
Xanthomonas vasicola pv. vasculorum NCPPB 1326

DUF4258

LCJM01000018.1:58-7534

Candidatus Giovannonibacteria bacterium GW2011_GWC2_44_8

? ? ? Acyltransferase ? ? DUF4258 DUF2283 Radical_SAM ?

KQ960127.1:2503-6181
Olsenella sp. DNF00959

Phage_min_cap2 DUF4258 ? ? Lin1244_N ? ? ? OPT ? ?

JACNLT010000075.1:328-3163
candidate division KSB1 bacterium

? DUF2283 DUF4258 ? ParE_toxin ?

JAAVHL010000040.1:19410-24962
Ruminococcus sp.

tRNA_m1G_MT RimM PRC_RimM DUF5104 KH_KhpA-B Ribosomal_S16 YxzE MqsA_antitoxin DUF4258 Imm17 SRP54_N SRP54 SRP_SPB UPF0122

JADGBI010000053.1:258-6377
SAR324 cluster bacterium

WGR DUF4258 MqsA_antitoxin His-Kinase Receiver S6PP SUI1 ? Porin_4

JADGAQ010000008.1:19535-35642
Magnetococcales bacterium

DUF3641 Radical_SAM Pyr_redox_dim Pyr_redox_2 Transketolase_C_1 Transket_pyr Transketolase_N SUI1 Sigma70_r4_2 Sigma70_r2 YfbK_C VWA vWF_A DUF2283 DUF4258 Trp_syntA PALP PSCyt1 NTP_transferase UDPG_MGDP_dh_C UDPG_MGDP_dh UDPG_MGDP_dh_N PGM_PMM_IV PGM_PMM_III PGM_PMM_II PGM_PMM_I DHO_dh

JADFXC010000003.1:42727-57348
Magnetococcales bacterium

eIF-1a OEP OEP SPOR MFS_1 FliG_C Phage_integrase DUF2064 MqsA_antitoxin DUF4258 LpxD Hexapep Hexapep Hexapep Hexapep Peptidase_S49 MerR_1 Enolase_N Enolase_C GDC-P GcvP2_C GDC-P GCV_T GCV_T_C

JADFWQ010000364.1:1-2450
Nitrospinota bacterium

HKD DUF2283 DUF4258 NtrC-AAA AAA_lid_14 HTH_8

JACYMB010000129.1:709-3589

Synechococcales cyanobacterium T60_A2020_003

DUF2283 DUF4258 DUF3368 UPF0175 LRAT ?

JACPCL010000038.1:12945-30927
Chloroflexi bacterium

Hydrolase_4 Hydrolase_4 EMP24_GP25L HTH_3 OCD_Mu_crystall MFS_3 Asp_protease_2 MqsA_antitoxin DUF4258 ? ILVD_EDD_N ILV_EDD_C CoA_transf_3 Anticodon_2 tRNA-synt_1c MTES_1575 AAA_12 AAA_11 AAA_11 Beta_elim_lyase MFS_3

JACRAQ010000103.1:28969-42044
Nitrosomonadales bacterium

PIN HD DEAD Cas3-like_C_2 ? Cas_Cas5d Cas_Csd1 Cas_Cas7 ? DUF4258 DUF2283 DDE_Tnp_1 DUF4372 DUF3368 Cas_Cas4 Cas_Cas1 CRISPR_Cas2

JACREW010000185.1:1-5374
Nitrospirae bacterium

Receiver DHQ_synthase DHQS_C DUF2283 DUF4258 DUF5615 DUF433 ParE_toxin Unstab_antitox SKI SNase Receiver CheB_methylest

JACREX010000138.1:299-5893
Bacteroidia bacterium

DUF4258 MqsA_antitoxin ? DUF2204 ? SLFN_AlbA_2 Trypsin PEGA 3keto-disac_hyd ABC2_membrane_2

JACRBW010000127.1:878-7635
Deltaproteobacteria bacterium

? MS_channel_1st_1 MS_channel_1st_1 TrkA_C TrkA_N Na_H_Exchanger HD HD_assoc ? QueC DUF2283 DUF4258 ? NTP_transf_2

JACRBV010000083.1:560-14637
Deltaproteobacteria bacterium

Pribosyl_synth Pribosyltran_N GHMP_kinases_C GHMP_kinases_N AAA_14 DUF4143 DUF4143 AAA_14 ? CYTH ClpB_D2-small AAA_2 AAA_lid_9 AAA Clp_N Clp_N DUF4258 DUF2283 Nramp CBS CBS MgtE_N PRC PDZ PDZ_2 Trypsin DUF1844 YfiO TPRs

WGLZ01000003.1:47741-5823

Cyanobacteria bacterium RI_101

DDE_Tnp_ISL3 DUF2283 DUF4258 ? PemK_toxin ? VWA ? Unstab_antitox FGS STY-kinase

JADIZD010000001.1:10355-21935
Saprospiraceae bacterium

AAA_14 DUF4143 ? HATPase_c_4 ABC2_membrane PIN ? DUF2283 DUF4258 Fic GATase_7 Asn_synthase PDDEXK_3 ? HepT-like Polbeta HicB_lk_antitox

JADKFU010000004.1:791865-810534

Chitinophagaceae bacterium

Complex1_30kDa Complex1_49kDa NADHdh Fer4_7 Oxidored_q3 Oxidored_q2 Proton_antipo_N Proton_antipo_M DUF4258 DUF2283 Proton_antipo_M Proton_antipo_M ? FAD_binding_6 NAD_binding_1 Por_Secre_tail Ig_7 PKD_4 PKD_4 PKD_4 Ig_7 PKD_4 ?

JADHVQ010000003.1:42967-55499
Bacteroidales bacterium

PAPS_reduct Mur_ligase_C Mur_ligase_M Mur_ligase ? PKD_5 Ig_7 PKD_4 ? Fib_succ_major DUF2283 DUF4258 Rho_N Rho_RNA_bind ATP-synt_ab DUF4293 RmuC SpoU_methylase SpoU_sub_bind ? ABC_tran OTCace_N OTCace

VGFB01000244.1:1-6257
Armatimonadetes bacterium

PfkB F_bP_aldolase ? ? DUF4258 DUF2283 FlgD_ig LYTB

JAFGIN010000073.1:1-3300
Chitinispirillaceae bacterium

? ? ? Polbeta HepT-like_2 DUF4258 MqsA_antitoxin

JAFGRC010000109.1:1-8996
Bacteroidales bacterium

Wzz GNVR ? DUF4160 DUF2283 DUF4258 Esterase AbiEii AbiEii DUF498 Mut7-C Ub-Mut7C ? SiaC ?

JAFLBZ010000035.1:898-7906
Rhodothermia bacterium

CarbopepD_reg_2 ? ? ? ? ? ? ? ? ? DDE_Tnp_IS66

JAGANQ010000013.1:20552-26511
Lachnospiraceae bacterium

Ribonuclease_3 tRNA-synt_1e DALR_2 CysS_C Hexapep YgbB ? RNR_inhib MqsA_antitoxin DUF4258 PSK_trans_fac Phage_integrase

JAGBML010000118.1:4034-12014
Selenomonas sp.

Fer4 Fe_hyd_lg_C Fe_hyd_SSU PIN ? Flg_bbr_C FlgE_D2 Resolvase Flg_bb_rod LlgE_F_G_D1 FlgE_D2 Flg_bbr_C DUF4258 ? Aldolase_II QueT Hydrolase_3 DUF815

JAGDCL010000234.1:1-6400
bacterium

TMP-TENI PBECR4 ADGRF3_N Lcl_C DUF4258 MqsA_antitoxin ?

JAFUYB010000028.1:37095-48157
Lachnospiraceae bacterium

ABC_tran ABC-3 AAA-ATPase_like PDDEXK_9 AAA-ATPase_like ? ? MqsA_antitoxin DUF4258 HTH_3 HTH_3 HTH_3 HTH_3 DUF2971 ? ? AbiGi

JAFVEU010000162.1:1257-4274
Lachnospiraceae bacterium

YokU DUF4258 ? DUF3881 DUF4446 ParB_N HTH_ParB ParB_C ParA-Soj-NTPase

JAFWVW010000045.1:2188-8819
Bacteroidales bacterium

PRK EamA EamA PDZ_6 Peptidase_S41 Ham1p_like RecO_C RecO_N ? ? DUF4258

JAFDXY010000009.1:402949-420484
Bacteroidetes bacterium

RlmN_N Radical_SAM S4 PseudoU_synth_2 PseudoU_synth_2 RecQ_Zn_bind Helicase_C DEAD Hydrolase E1-E2_ATPase HMA TsaE AlaDh_PNT_N AlaDh_PNT_C DUF4258 DUF2283 Peptidase_S41 PDZ_Tricorn Tricorn_C1 Beta-prop_THOC3 PD40 PD40 PD40 ? RuvB_N AAA_lid_4 RuvB_C Ubie_methyltran OMP_b-brl_2 ?

JAGVWD010000003.1:3761-14103
Candidatus Iainarchaeum archaeon

? Peptidase_M24 PEP-utilizers ? HD ? ? DUF4258 DUF2283 Na_H_Exchanger CopD

JAGZZB010000092.1:4858-6171
Clostridiales bacterium

HicB_lk_antitox E6 DUF4258 ? DUF7666 HicB_lk_antitox Phage_min_cap2 ? DUF6275 cwf21

JAHBTQ010000002.1:592975-606028
Taibaiella sp.

Mur_ligase_C Mur_ligase_M Mur_ligase PMT_2 Glycos_transf_2 YkuD ? Pyr_redox_2 ? DUF4258 DUF2283 ? ? Terminase_5 ? Glyoxalase Bac_luciferase

JADLHL010000169.1:1-10285

Chitinophagaceae bacterium

? ABC_tran ABC_tran_Xtn ABC_tran HTH_3 Fic Porin_O_P Ala_racemase_N DUF4258 Radical_SAM UPF0004 BCCT Methyltransf_24 DHFR_1 Thymidylat_synt

JADZCH010000111.1:443-4143
Ignavibacteria bacterium

GIY-YIG RNase_T TPMT DUF4258 DUF2283 ?

JAJBTC010000008.1:2464-18983
Bacteroides sp.

DUF3108 4HBT Glyco_hydro_88 Sulfatase His-Kinase DUF3836 RHS_repeat DUF6443 ? ? SGTA_dimer ? ? ? ?

JAGHCD010000454.1:2001-10819
Spirochaetaceae bacterium

FtsX MacB_PCD FtsX LolA_like ? ABC_tran NTP_transferase dTDP_sugar_isom DUF4258 MqsA_antitoxin RmlD_sub_bind

WANR01000055.1:204-10616
Desulfurococcales archaeon

RAMPs ? ? DUF3368 UPF0175 ? TumE DUF4258 DUF2283 ? AF2212-like ? ? Acetylase Polbeta

WAJZ01000018.1:1092-9337
Candidatus Hestiella acidicharens

Radical_SAM SNO ThiG SOR_SNZ ? ? AF2212-like DUF2283 DUF4258 AF2212-like Memo ? adh_short_C2 Slo-like_RCK Ion_trans_2

JAKISR010000098.1:585-2147
Sulfurimonas sp.

DUF4258 YokU DNA_repr_REX1B MqsA_antitoxin

JAIPDI010000021.1:73966-86689
Deltaproteobacteria bacterium

Methyltransf_25 HicA_toxin HicB ORF6N ? ParE-like_toxin Radical_SAM DUF4258 DUF2283 DNA_processg_A

JAIPAB010000074.1:12989-21752
Melioribacteraceae bacterium

DegT_DnrJ_EryC1 GDP_Man_Dehyd Epimerase Epimerase 23S_rRNA_IVP Epimerase MqsA_antitoxin DUF4258 DUF2283 DUF2283 GDP_Man_Dehyd ? PIN ? DUF4258

JADFIH010000078.1:1071-12831
Proteobacteria bacterium

tRNA-synt_2c tRNA_SAD PALP His-Kinase His-Kinase PAS PAS PAS PAS MlaC ? Rhodanese MqsA_antitoxin DUF4258 ? Abhydrolase_1 FAD_binding_2 DctP

JACZRX010000002.1:18724-27671
Chloroflexota bacterium

? ? ? ? Phage_capsid Peptidase_S78 Fe-S_biosyn DUF4258 MqsA_antitoxin G_glu_transpept ? HNH ? Arch_flagellin

JALFEB010000026.1:19289-29271
Oscillospiraceae bacterium

DUF6291 DnaB_2 ? ? DUF1492 HicB_lk_antitox HNH ? DUF4258 ? ? DUF4355 Phage_capsid ? Phage_Gp19 ?

JAGLIC010000106.1:88-1817
Bacteroidales bacterium

DUF4258 YokU T5orf172 Sel1

JALMSA010000098.1:29237-51037
Comamonadaceae bacterium

? EAL GGDEF PAS PAS NAD_binding_1 FAD_binding_6 PNTB_4TM AlaDh_PNT_N AlaDh_PNT_C CorA DUF2283 DUF4258 ? ? MmeI_Mtase PP2C_2 FAD_binding_4 STAS Sulfate_transp ?

JALNZD010000045.1:8884-22196
Bacteroidales bacterium

GH97_N Glyco_hydro_97 GH97_C ? tRNA-synt_2d Phe_tRNA-synt_N Pro_isomerase TK ATPase_2 DUF5112 His_kinase His-Kinase DUF4258 DUF2283 ? TP_methylase ? NUDIX LuxE LuxE ?

WRJE01000046.1:1-8424
Oscillospiraceae bacterium

S4 DUF402 ParE_toxin PhdYeFM_antitox DXP_synthase_N Transket_pyr Transketolase_C MqsA_antitoxin DUF4258 DUF212 polyprenyl_synt Exonuc_VII_S tRNA_anti_2 Exonuc_VII_L Exonuc_VII_L TsaD TsaD NusB Asp23

JAMCPS010000022.1:1594-17637
Epsilonproteobacteria bacterium

TruB_N PcrA_UvrD_tudor UvrD_C UvrD-helicase HTH_1 LysR_substrate ? Aminotran_4 Aminotran_4 PALP Biotin_carb_N CPSase_L_D2 Biotin_carb_C DUF4258 MqsA_antitoxin McrBC CN_hydrolase YajC SecD_1st SecDF_P1_head SecD_SecF_C Sec_GG SecD_SecF_C tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 Anticodon_1

JAGFJE010000430.1:17191-27004
Phaeodactylibacter sp.

? HTH_18 ? Amino_oxidase Amino_oxidase MlaE ABC_tran MlaD DUF4258 DUF2283 HTH_18 Beta-lactamase ABC_tran

JACKOD010000001.1:736872-758409
Hahellaceae bacterium

? DUF927 Zn_Ribbon_Prim DUF7146 Toprim_3 ? Phage_AlpA HlyD_D4 ? MqsA_antitoxin DUF4258 Arm-DNA-bind_3 Phage_int_M Phage_integrase Phage_integrase Phage_integrase ? MZB DUF7130 Helicase_C DEAD GFA Trigger_C FKBP_C Trigger_N

JAKSKX010000016.1:17824-32767
Paludibacteraceae bacterium

FKBP_C DHH DHHA1 UPF0029 DEAD Helicase_C DbpA ? ? ? DUF4258 DUF1015 Ubie_methyltran COesterase ? DUF4377 Hydrolase_3 Thioredoxin Glycos_transf_N

QMIT01000003.1:338632-356576
Chloroflexota bacterium

Biotin_lipoyl_2 OEP HlyD_3 ABC_tran ABC_tran LysM HAMP His-Kinase His-Kinase Receiver Trans_reg_C Acetylase DUF4258 DUF2283 2-oxoacid_dh E3_binding Biotin_lipoyl Transketolase_C Transket_pyr E1_dh ? STY-kinase nSTAND1 Beta-prop_THOC3 WD40_Prp19 Beta-prop_WDR3_1st WD40_CDC20-Fz ?

JAILFL010000123.1:1-17086
Clostridia bacterium

ABC_membrane ABC_tran ? GATase_7 Asn_synthase NIF ? GATase_7 Asn_synthase Smr MSSS MutS_V DUF4258 YokU RimP_N DUF150_C NusA_N S1 KH_5 KH_NusA_C YlxR IF2_N IF2_N GTP_EFTU EF-G_D2 IF-2 GTP_EFTU_D2 RBFA DHH DHHA1

JANWYN010000117.1:19135-10107
Candidatus Roseilinea sp.

? ? DUF4258 DUF2283 Na_Ca_ex Na_Ca_ex Na_Ca_ex Na_Ca_ex Proton_antipo_N Proton_antipo_M MbhD MbhE MnhB Oxidored_q2 Proton_antipo_M MrpF_PhaF

JANWUY010000223.1:137-2904
Bacteroidia bacterium

FA_desaturase LRRs LRR_14 Rieske DUF4258 Peptidase_M16

JAMXFE010000004.1:279555-310505
Laspinema palackyanum D2b

His-Kinase Receiver EAL GGDEF Receiver His-Kinase His-Kinase Receiver His-Kinase HisKA_2 PAS PAS PAS PAS PAS PAS His-Kinase His-Kinase PAS Receiver BPD_transp_1 DUF2283 DUF4258 ? ? Sigma70_r2 Sigma70_r4_2 TamB DUF3110 GKRP-like_C SIS GKRP_SIS_N RuBisCo_chap_C Raf1_N Raf1_HTH

JAOZHW010000022.1:73776-89947
Candidatus Bathyarchaeota archaeon

Gln-synt_N Gln-synt_C pre-PUA_NIP7 UPF0113 Methyltr_RsmF_N Methyltr_RsmB-F TrkH TrkH NikR_C RHH_1 NadA DUF2283 DUF4258 DUF5915 Anticodon_1 tRNA-synt_1 C2HC_ASCIZ ATP-grasp_5 PilB3_C T2SSE T2SSF T2SSF ? DUF7266

JAPLDH010000019.1:65959-86878
Bacteroidetes bacterium

CotH HTH_3 Peptidase_S24 IMS_C IMS_HHH IMS HHH_6 DNA_pol3_finger DNA_pol3_alpha PHP 23S_rRNA_IVP Fib_succ_major IgGFc_binding DUF2283 DUF4258 Spermine_synth Dimerisation Methyltransf_2 Band-7 ? Peptidase_M16 Peptidase_M16 Peptidase_M16_C Peptidase_M16_C PSII_BNR PSII_BNR PSII_BNR PSII_BNR Fib_succ_major

JAPMTR010000105.1:656-1768
Candidatus Methanoperedens sp.

MqsA_antitoxin DUF4258 ABC_tran

JAPULP010000424.1:1-3367
Gemmatimonadetes bacterium

Beta-prop_THOC3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 MqsA_antitoxin DUF4258

JAQTZZ010000025.1:5263-17098
Candidatus Wallbacteria bacterium

? Secretin Secretin_N Secretin_N STN ? CdsD_C PilO PilN DUF4143 AAA_14 DUF4258 DUF2283 HTH_3 HTH_3 Gp49 ? C1q Peptidase_S74

JARRBC010000002.1:34653-46668
Thermococcaceae archaeon

? ? HMG-CoA_red CbiA ADH_N ADH_zinc_N ? TrmB tRNA-synt_2d DUF4258 DUF2283 ? PIN PhetRS_B1 B3_4 B5 tRNA_synthFbeta ? PseudoU_synth_1 PseudoU_synth_1 Helicase_C ResIII

JASEFY010000192.1:1206-1873
Candidatus Bathyarchaeia archaeon

Glycos_transf_4 DUF4258 DUF2283

JASJCT010000013.1:19844-45499
Myxococcota bacterium

DUF6599 N_methyl N_methyl N_methyl PilX_N T4P_PilY1 ? DUF4258 MqsA_antitoxin RF-1 PCRF DUF1385 Ribosomal_L31 ATP-synt_ab Rho_RNA_bind Rho_N ? TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs

JAUDBU010000005.1:64341-75074
Phascolarctobacterium faecium

Sua5_yciO_yrdC Sua5_C PrmC_N MTS PCRF RF-1 DUF1385 Ribosomal_L31 2Fe-2S_thioredx MqsA_antitoxin DUF4258 Asn_synthase FlK ABC_tran BPD_transp_1 SBP_bac_3 Cupin_2 LysE

JAUSCP010000168.1:285-3220
Gallionella sp.

DUF4258 DUF2283 DUF4347 PIN ? ? ? BrnT_toxin

JAUZRY010000101.1:5366-16766
Ardenticatenia bacterium

? DUF3592 Beta_helix Beta_helix DDE_Tnp_IS1 HisKA_3 His-Kinase Receiver GerE ? DUF4258 DUF2283 gag-asp_proteas Asp_protease_2 UPF0175 DUF3368

JAIROQ010000214.1:1-3363
Prevotellaceae bacterium

Acyl-CoA_dh_N Asp_protease_2 ? ETF ETF_alpha DUF2283 DUF4258 ETF

JAXHUZ010000318.1:403-3000
Thermodesulfobacteriota bacterium

DUF4258 MqsA_antitoxin ? ? PUA AA_kinase

JAXHMA010000003.1:547155-557817
Balneolaceae bacterium

? Hydrolase_3 Transposase_20 DEDD_Tnp_IS110 DUF4258 YokU Unstab_antitox ParE_toxin DUF4258 MqsA_antitoxin ? DUF433 ? DUF2283 DUF433 DUF5615

JAYEJM010000027.1:430-8658
Methanobacteriota archaeon

HicA_toxin UPF0150 UPF0150 HicB_lk_antitox DUF4258 MqsA_antitoxin Mur_ligase_M oligo_HPY ABC_tran oligo_HPY ABC_tran BPD_transp_1 OppC_N BPD_transp_1 BPD_transp_1_N Methyltransf_2 Dimerisation

WWDT01000080.1:3610-4402
Candidatus Tiamatella sp.

DUF2283 DUF4258 ? ? ? DUF3368 UPF0175 ThiF

VXSR01000046.1:5418-18109
Gemmatimonadetes bacterium

HTH_37 Abhydrolase_11 MOSC DNA_binding_1 NUDIX NUDIX 3keto-disac_hyd HTH_3 AstE_AspA_cat DUF4258 DUF2283 DUF5683 DUF5683 SGSH_C Sulfatase Arylsulfotrans Arylsulfotrans PhyH SGSH_C Sulfatase SGSH_C Sulfatase

VXOG01000034.1:85246-97701
Chloroflexi bacterium

DUF772 DDE_Tnp_1 Apolipoprotein ? ? ? Phe_tRNA-synt_N tRNA-synt_2d ? DUF4258 DUF2283 tRNA_bind B3_4 B5 tRNA_synthFbeta FDX-ACB MoeA_N MoCF_biosynth MoeA_C ? IF3_N IF3_C Ribosomal_L35p Ribosomal_L20

JAAAOL010000245.1:1-4641
Bacteroidetes bacterium

? BPD_transp_1 DUF2283 DUF4258 BPD_transp_1 SBP_bac_8

JAAYFG010000030.1:16507-3359
Tissierellia bacterium

DUF4258 ? ? Acb2_Tad1_hairpin DUF4355 ? Phage_connect_1 ?

JAADIZ010000158.1:1335-3158
Campylobacterota bacterium

DUF772 YCII DUF4258 YokU ? ? LRRs

JABXKB010000332.1:137-4954
Promethearchaeota archaeon

NTP_transf_2 DUF2031 PIN Na_Ca_ex Na_Ca_ex DUF2283 DUF4258 ADH_zinc_N_2 AlaDh_PNT_C ADH_N

JACAEL010000028.1:1741-13612
Candidatus Thorarchaeota archaeon

? SLFN_AlbA_2 HicA_toxin UPF0150 ? PIN NYN DUF4258 DUF2283 ? HepT-like ? ? PIN DUF2281

MEOZ01000178.1:1349-19231

Bacteroidetes bacterium RIFCSPLOWO2_02_FULL_36_8

Usp Usp Na_H_Exchanger PKD_4 CHU_C UPF0102 PKD_4 Por_Secre_tail DUF6922 YokU DUF4258 ? PNGase_F-II_N ? Por_Secre_tail Reeler

MFHN01000005.1:20546-32042

Candidatus Giovannonibacteria bacterium RIFCSPHIGHO2_02_FULL_44_51

? ? DUF5649 PilM_2 ? ? ? DUF4258 DUF2283 ? ? HSP20 PrsW-protease Anticodon_1 tRNA-synt_1 tRNA-synt_1 TsaD

MGDS01000050.1:34885-50010

Candidatus Uhrbacteria bacterium RIFCSPHIGHO2_01_FULL_47_11

KH_KhpA-B ? DUF4143 AAA_14 RHH_1 PIN ? ? DUF4258 DUF2283 ? ? 23S_rRNA_IVP LAGLIDADG_1 LAGLIDADG_2 DHH DHH DHHA1

PEWM01000238.1:1575-5921

Deltaproteobacteria bacterium CG07_land_8_20_14_0_80_60_11

? DUF2283 DUF4258 ATP-synt_DE_N ATP-synt_DE ATP-synt_ab_N ATP-synt_ab ATP-synt_VA_C ATP-synt ATP-synt_ab_C

PSRN01000059.1:1-5258
Bdellovibrio sp.

Peptidase_S8 DUF4258 ? ? DUF4143 ? PYNP_C Glycos_transf_3 Glycos_trans_3N ? DeoC

QGTG01000013.1:61867-78814
Chitinophaga sp. S165

FeoA Fe_dep_repr_C Fe_dep_repress ? Leu_Phe_trans ClpS Na_H_Exchanger Usp Glyco_transf_28 EryCIII-like_C Radical_SAM UPF0004 DUF4258 Acyltransferase PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C NAD_binding_10 FAD_binding_4 MurB_C Pyr_redox_3 GATase LCM

QULB01000004.1:131136-140526
Firmicutes bacterium OM07-11

DUF7336 ? ? ? ? GP3_package YokU DUF4258 ? ? ? ? MazG YopX ?

QNBQ01000223.1:1-4468
Candidatus Poribacteria bacterium

DegT_DnrJ_EryC1 DUF1858 SATase_N Hexapep Hexapep Lactamase_B 2-Hacid_dh_C PGDH_inter ACT YokU DUF4258 HD_assoc

QMLI01000232.1:76-2645
Deltaproteobacteria bacterium

PIN DUF4258 MqsA_antitoxin N6_N4_Mtase

QMPI01000069.1:1-9440
Bacteroidetes bacterium

Amino_oxidase Citrate_synt DUF4258 AMP-binding AMP-binding_C_3 Acetylase DUF6600 CTP_synth_N MqsA_antitoxin

QMYR01000137.1:1519-6918
Candidatus Thorarchaeota archaeon

? ? ? ? ? DUF2283 DUF4258 ? Acetylase

RFIF01000227.1:3379-8114
Cyanobacteria bacterium J069

Aldedh ? DUF2283 ? Big OpuAC Uma2 DUF4258 ?

FMGQ01000007.1:20500-29905
uncultured Clostridium sp.

HNH_3 ? ? DUF6906 ? HNH SPP1_portal VG15 DUF4258 ? ? ? ? Phage_capsid Phage_Gp19 ?

FMFL01000058.1:14883-20835
uncultured Ruminococcus sp.

? IMS IMS_HHH IMS_C Calcineurin ? DUF7675 Transglutaminase ? DUF4258

RDYG01000257.1:1-3752
Runella slithyformis

Pribosyl_synth Ribosomal_L25p Ribosomal_TL5_C RBFA DUF4258 DUF2283 CM_2 PDT ACT Thymidylat_synt

SCUG01000536.1:1-2685
Saprospiraceae bacterium

DUF4258 DUF433 AAA-ATPase_like PDDEXK_9 GNVR

SMBS01000004.1:261688-272310
Tissierella praeacuta

? ? ? ? Phage_scaffold Phage_gp49_66 ? Phage_min_cap2 Phage_min_cap2 ? ? HTH_Tnp_1 ? ? ? MethyltransfD12

VFMD01000368.1:1-2284
Halothiobacillaceae bacterium

HrcA HrcA_DNA-bdg NAD_kinase NAD_kinase_C DUF4258 DUF2283

CABWIG010000011.1:255887-265832
Collinsella aerofaciens

Minor_capsid_2 Minor_capsid_1 ? ? ? Phage_scaffold ? Phage_min_cap2 SPP1_portal ? Terminase_3 ? ? ? ?

NZ_JAWYAU010000285.1:1-10172
Enterocloster asparagiformis

Peripla_BP_3 LacI ABC_tran ABC_tran SBP_bac_1 MqsA_antitoxin DUF4258 BPD_transp_2 ROK Asn_synthase Peripla_BP_4 Peripla_BP_4

NZ_ALNK01000021.1:10040-19504
Peptoanaerobacter stomatis

DUF6261 NikA-like ? SPP1_portal GP7 Phage_min_cap2 DUF4258 Phage_scaffold Rho_N ? DUF6751 DUF6751 DUF1492 ? Terminase_2 ? Terminase_6N

NC_014721.1:734270-741997
Caldicellulosiruptor acetigenus I77R1B

? ? RHH_1 ? DUF4258 YokU Bro-N ANT PemK_toxin DUF7387 ? PemK_toxin ? RHH_1 ? HicA_toxin

NC_015681.1:2026192-2042900
Thermodesulfatator indicus DSM 15286

? F_T4SS_TraN ? TraC_F_IV P-loop_TraG DUF4258 ArdcN MPTase-PolyVal DUF3782 DUF736 YdjM HD TraG-D_C PIN HNH_repeat His-Kinase MqsA_antitoxin TrbC_Ftype

NC_015931.1:285373-299500
Pyrolobus fumarii 1A

? ? DUF4258 ? Phage_integr_3 RHH_1 ? Zot HerA-FtsK ? ODP Flavodoxin_1 ? DUF2283 ? ? zf-C2H2

NZ_CAUCJO010000022.1:12308-28198
uncultured Ruminococcus sp.

BPD_transp_1 Sortase BPD_transp_1 crAss001_48 DUF1284 HTH_3 HTH_3 LRRs LRRs LRRs LRRs LRRs LRRs Gram_pos_anchor ? DUF4258 PfkB Glyco_hydro_32N Glyco_hydro_32C ABC_tran MIT_C OB_MalK TOBE_2 SBP_bac_8 Peripla_BP_3 LacI His-Kinase His-Kinase HAMP

NZ_KI669560.1:329139-341566
Sediminibacterium sp. C3

Phe_tRNA-synt_N tRNA-synt_2d DUF4290 EPSP_synthase AbiEi_2 ? ? HTH_Tnp_1 HTH_Tnp_1 ZMYND8_CC ? ? RecO_N RecO_C HTH_21 rve rve_3 ? AbiEii

NZ_DILG01000061.1:281-10236
Sulfuricurvum sp. UBA5598

DUF234 IMS_C IMS_HHH IMS ? Aldo_ket_red Aminotran_1_2 DUF4258 MqsA_antitoxin KAP_NTPase KAP_NTPase AP_endonuc_2 ? Vps55

NZ_URZQ01000077.1:6322-26490
uncultured Bacteroides sp.

DUF2283 DUF4010 MgtC ? OMP_b-brl_3 Plug CarbopepD_reg_2 RF-1 PCRF Pect-lyase_RHGA_epim Glyco_hydro_28 CYTH Fn3-like Glyco_hydro_3_C Glyco_hydro_3 Peptidase_M16 Peptidase_M16 Peptidase_M16_C Peptidase_M16_C ? ? DUF4258 Calcineurin PQQ_2 Beta-prop_ACSF4 PQQ_2 GCE_fung

NZ_AP019796.1:2085812-2100960
Rhodothermus marinus

MqsA_antitoxin RAMPs Cas_Cmr3 Cmr2_N Cas10-Cmr2_palm2 RAMPs Cas_Cas4 Cas3_HD DEAD Cas3-like_C_2 DUF4258 Cas_Cas5d DevR ? Cas_Cas6_C HTH_11 WYL WCX HepT-like_2 NTP_transf_2

NZ_BBBE01000004.1:3919-20010
Thermococcus peptonophilus OG-1 = JCM 9653

TRM5-TYW2_MTfase DUF4258 ? DUF11 DHH DHHA1 Glycos_transf_2 ? Methyltransf_21 TauE TauE DUF2283 ATP-sulfurylase PUA_2 DUF7344 ?

NZ_LN898210.1:323216-336570
Olsenella massiliensis

ABC_tran Phage_scaffold YjcQ ? Phage_Mu_F DUF4258 Zn_ribbon_DnaG DNAG_N Toprim_2 DnaB_bind FAD_syn Flavokinase TruB_N TruB_C_2 DHH DHHA1 RBFA Glycos_transf_2 HK97-gp10_like Sigma70_r1_1 Sigma70_r1_2 Sigma70_r2 Sigma70_r3 Sigma70_r4 HD_4 ABC2_membrane

NZ_BSDW01000001.1:2103196-2119240
Tepidanaerobacter syntrophicus

Put_DNA-bind_N CoA_binding MqsA_antitoxin DUF4258 DUF1664 NTP_transferase ManC_GMP_beta-helix HTH_30 GGDEF_2 TOBE OB_MalK ABC_tran PGA_cap Peptidase_S26 Transposase_31 DUF4351 UPF0236 MMR_HSR1 YchF-GTPase_C Germane Germane Cu_amine_oxidN1 AMIN Amidase_3 ?

NZ_CAMYEL010000015.1:3602-20196
Olegusella massiliensis

MscS_porin CC_PcsB NLPC_P60 CRISPR_Cse1 CRISPR_Cse2 Cas_CT1975 Cas_Cas5d CRISPR_assoc Cas_Cas1 Cas_Cas1 Cas_Cas2CT1978 RNase_T Phage_min_cap2 DUF4258 ? TMF_TATA_bd ? GDPD MurJ ATPgrasp_Ter

NZ_PYCI01000099.1:8694-22890
Chroococcidiopsis sp. CCALA 051

Lactonase Big Big_13 Big_13 Big_13 Big_13 NikA-like Bro-N CEP209_CC5 DUF4258 ParA-Soj-NTPase ? DUF3368 UPF0175 ? Relaxase ? DUF2283 ?

NZ_QPIZ01000016.1:10696-26934
Marinilabilia salmonicolor

CBM96 Por_Secre_tail ? DUF4258 ? MazE_antitoxin Calcineurin Zn_ribbon_Nudix DUF4852 Phage_integrase Phage_int_SAM_5 Arm-DNA-bind_5 ? DUF4279 ? SpvB FG-GAP_3 FG-GAP_3 RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat DUF6557 Ntox33

NZ_CP127527.1:134935-141010
Acidithiobacillus sulfuriphilus

HTH_23 HTH_33 HsdM_N N6_Mtase BrnA_antitoxin ? DUF4258 MqsA_antitoxin HTH_37 ? PemK_toxin DUF4926 BrnT_toxin Thioredoxin_3 ? ORF6N DDE_3

NZ_CP035945.1:2510298-2529253
Blautia producta

Alpha_L_fucos SBP_bac_1 PHP Receiver GGDEF_2 HTH_18 ? DUF4258 AraC_binding HTH_18 GFO_IDH_MocA Glyco_hydro_109_C SEC-C SWIM Big_15 SNF2_assoc SNF2-rel_dom Helicase_C Plexin_RBD DUF218 Alpha_L_fucos HAMP His_kinase His-Kinase YokU

NZ_VNHY01000001.1:1020622-1038215
Fodinibius salinus

? ? Sulfotransfer_3 MqsA_antitoxin GTP_EFTU GTP-eEF1A_C PAPS_reduct APS_kinase 23S_rRNA_IVP Inositol_P CitMHS CitMHS HHH_3 Sulfotransfer_1 Polysacc_synt MatE MurJ Wzz DUF4258

NZ_AZNR01000018.1:16652-9473
Acidithiobacillus ferrooxidans BY-3

GNVR HTH_24 DUF4258 MqsA_antitoxin Epimerase Polysacc_synt Glycos_transf_1 Glyco_transf_4 Glycos_transf_1 ? Glycos_transf_2

NZ_CP046565.1:1597676-1611355
Methylococcus geothermalis

rve_3 rve HTH_21 HTH_Tnp_1 AAA_lid_2 AAA_3 ? ? FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 ? ? Arm-DNA-bind_3 Phage_int_M Phage_integrase DUF4258 YokU HTH_3 UvrD_C UvrD-helicase Mg_chelatase_C Mg_chelatase ChlI P-II BMFP

NZ_JALIFP010000003.1:22675-32907
Lederbergia sp. NSJ-179

SNF2-rel_dom Helicase_C Sigma70_r4_2 ? HTH_Tnp_1_2 Terminase_6N Terminase-T7_RNaseH-like IcmW SPP1_portal Phage_Mu_F ? ? Tad2-like Phage_scaffold Phage_capsid_4 ? Phage_HCP

NZ_JACRSU010000002.1:316596-329730
Congzhengia minquanensis

? ? ? ? Phage_scaffold ? ? Phage_min_cap2 DUF4258 ? Terminase_6N DUF1492 SNF2-rel_dom VRR_NUC VapE-like_dom HTH_23

NZ_JAKKUT010000002.1:1772467-1785470
Candidatus Synechococcus calcipolaris G9

ParE_toxin ? CRISPR_assoc Cas_Cas5d Cas_CT1975 HicB_lk_antitox DUF2283 DUF4258 CRISPR_Cse2 CRISPR_Cse1 Cas3_HD DEAD Cas3-like_C_2 Cas3_C HTH_11 WYL WCX RNase_T Calcineurin Acetylase

NZ_JARXWF010000021.1:1866-15058
Parabacteroides sp. PM6-13

? ? ? ? ? RHS DUF4258 ? DUF4258 DUF4332 tRNA-synt_2 N-OB NAD_Gly3P_dh_C NAD_Gly3P_dh_N PGI PP_kinase_N PP_kinase PP_kinase_C_1 PP_kinase_C GCE_fung ?

NZ_CALJJL010000021.1:23822-33736
uncultured Treponema sp.

? ? DnaT_2 ? P22_CoatProtein ? ? Phage_Mu_F DUF4258 ? ? DUF4417 ADPrib_exo_Tox DUF4055 Terminase_6N DUF5681

NZ_CALFZW010000050.1:1-4102
uncultured Mesotoga sp.

? ? DUF4258 Pribosyltran DNA_processg_A

NZ_WFPS01000088.1:16545-9775
Persephonella sp.

Bac_DNA_binding Hexapep_2 Hexapep Hexapep PTPS YgjP-like Methylase_S ? ? DUF2283 DUF4258 REase SWI2_SNF2 T1R_D3-like T1RH-like_C Rad52_Rad22 LpxB PIN

NZ_CAUDTN010000005.1:4280-14602
Senegalimassilia anaerobia

Phage_scaffold ? ? SU10_MCP HK97-gp10_like VRR_NUC ? Phage_Mu_F DUF4258 SPP1_portal Terminase_3 ? Radical_SAM ParB_N ? ?

NZ_CAMJRK010000018.1:8543-8121
uncultured Leuconostoc sp.

HTH_23 Terminase_2 SPP1_portal Phage_Mu_F DUF4258 ? Phage_scaffold Gp13-like HeH ? ? HK97-gp10_like Terminase_6N Terminase_6C

NZ_CALACG010000160.1:1-3851
uncultured Campylobacter sp.

? Prok-RING_1 ? ? DUF4258

NZ_JAQXWZ010000094.1:11623-21413
Anaerovibrio sp.

IBR_1 ? Terminase_2 Terminase_3 Terminase_3C SPP1_portal ? ? Phage_Mu_F ? DUF2513 ? DUF4355 Phage_capsid ? ?

NZ_JARIDN010000281.1:440-1722
Clostridium sp.

DUF2513 ? ? DUF4258

NZ_JBNARD010000116.1:289-775
Methanothrix sp.

DUF4258 DUF2283